



## R1.2 Energy cycles in reactions (HL)

### Reactivity 1 · What drives chemical reactions? · IB Diploma · Concept guide and original exam-style practice

#### Understand the concept

Because enthalpy is a state function, Hess's law allows enthalpy changes that cannot be measured directly to be calculated by combining measurable/known enthalpy changes into a cycle, such as an energy cycle using enthalpies of formation or combustion, or a Born-Haber cycle for lattice enthalpy of an ionic compound (combining ionization energies, electron affinities, atomisation/sublimation enthalpies, bond enthalpy, and enthalpy of formation). Average bond enthalpies can also be used to estimate the  $\Delta H$  of a reaction by considering the energy required to break bonds in reactants (endothermic) minus the energy released forming bonds in products (exothermic), though this gives only an approximate value since average bond enthalpies vary with molecular environment.

#### Worked example: follow the reasoning

Use a Born-Haber cycle to calculate the lattice enthalpy of formation of sodium chloride,  $\text{NaCl(s)}$ , given:  $\Delta H_f^\circ(\text{NaCl}) = -411 \text{ kJ mol}^{-1}$ ; enthalpy of atomisation of  $\text{Na(s)} = +107 \text{ kJ mol}^{-1}$ ; first ionization energy of  $\text{Na(g)} = +496 \text{ kJ mol}^{-1}$ ; enthalpy of atomisation of  $\text{Cl}_2(\text{g}) = +122 \text{ kJ mol}^{-1}$  (per mole Cl atoms); first electron affinity of  $\text{Cl(g)} = -349 \text{ kJ mol}^{-1}$ .

Step 1: Write the cycle:  $\Delta H_f^\circ(\text{NaCl}) = \Delta H_{\text{atom}}(\text{Na}) + \text{IE1}(\text{Na}) + \Delta H_{\text{atom}}(\text{Cl}) + \text{EA1}(\text{Cl}) + \Delta H_{\text{lattice}}(\text{NaCl})$ .

Step 2: Substitute known values:  $-411 = (+107) + (+496) + (+122) + (-349) + \Delta H_{\text{lattice}}$ .

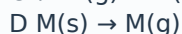
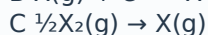
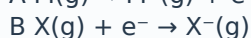
Step 3: Sum known terms:  $107 + 496 + 122 - 349 = 376$ .

Step 4:  $-411 = 376 + \Delta H_{\text{lattice}} \rightarrow \Delta H_{\text{lattice}} = -411 - 376 = -787 \text{ kJ mol}^{-1}$ .

Answer: Lattice enthalpy of formation of  $\text{NaCl} = -787 \text{ kJ mol}^{-1}$ .

#### Quick check · multiple choice

In a Born-Haber cycle for an ionic compound  $\text{MX}$ , which step corresponds to the first electron affinity of  $\text{X}$ ?



Write your choice: \_\_\_\_\_

### Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Use the following average bond enthalpy data to estimate the enthalpy change for the reaction: [6]
- $$\text{CH}_4(\text{g}) + 3\text{Cl}_2(\text{g}) \rightarrow \text{CHCl}_3(\text{g}) + 3\text{HCl}(\text{g})$$
- Bond enthalpies ( $\text{kJ mol}^{-1}$ ): C-H = 414, Cl-Cl = 242, C-Cl = 338, H-Cl = 431.
- (a) State the number and type of bonds broken and formed in this reaction. [2]
- (b) Calculate the enthalpy change of the reaction using bond enthalpies. [3]
- (c) Explain why a value calculated using average bond enthalpies is only an approximation of the true enthalpy change. [1]

Final answer: \_\_\_\_\_



## Answers and explanation

### Question 1 [6]

(a) Bonds broken:  $3 \times \text{C-H}$ ,  $3 \times \text{Cl-Cl}$ . Bonds formed:  $3 \times \text{C-Cl}$ ,  $3 \times \text{H-Cl}$  (one C-H bond remains unbroken in the product  $\text{CHCl}_3$ ). [2]

(b) Energy to break bonds (endothermic, +):  $3(414) + 3(242) = 1242 + 726 = 1968 \text{ kJ}$ . Energy released forming bonds (exothermic, -):  $3(338) + 3(431) = 1014 + 1293 = 2307 \text{ kJ}$ .  $\Delta H = \text{bonds broken} - \text{bonds formed} = 1968 - 2307 = -339 \text{ kJ mol}^{-1}$ . [3]

(c) Average bond enthalpies are averaged over many different compounds/molecular environments, so the actual bond enthalpy for a specific bond in this particular molecule may differ from the average value used, making the calculated  $\Delta H$  approximate. [1]

### Quick check answer

B. The first electron affinity is the enthalpy change when one mole of gaseous atoms of X each gain one electron to form gaseous  $\text{X}^-$  ions.

#### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.